The Riemann Hypothesis in the PEN Universe: Synthetic Prime-Trace Operators and the Critical Line

Halvor Lande

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Abstract

We work inside the fully coupled univalent PEN universe at steps 14 and 15, where the selected library already contains inner products, orthogonal decomposition, spectral calculus for self-adjoint operators, cohesion, and the temporal-cohesive shell. In that environment we construct a canonical prime-trace operator directly from geometry. The carrier is the synthetic idele-class space

$$X_\zeta := \mathbb{A}^\times/\mathbb{Q}^\times,$$

assembled top-down from the geometric circle, its loop group, Stone completion, the derived real line, and quotient formation already available in the library. The step-14 Hilbert package yields the critical Hilbert space

$$V_\zeta \subseteq L^2(X_\zeta),$$

while the step-15 temporal shell internalizes the scaling flow of $\mathbb{R}_{>0}$ as a strongly continuous unitary action. Its Stone generator

$$H_\zeta = -i\nabla_{v_{\mathrm{scale}}}\big|_{V_\zeta}$$

is self-adjoint.

The main analytic theorem proves that the compressed scaling action satisfies the exact Weil explicit formula:

$$\mathrm{Tr}\Big(U(h)\big|_{V_\zeta}\Big) = W(h)$$

for every compactly supported smooth multiplicative test function h, where W is the standard Weil distribution. Because the action is unitary and the step-14 cutoff is orthogonal and invariant, one obtains

$$W(g * g^*) \geq 0$$

for every test function g. By Weil's criterion, the Riemann Hypothesis follows. We then show that the same operator is also the unique PEN-optimal realization of the prime-trace API: any off-line branch requires at least three extra MBTT clause families—pairing, reflection, and non-self-adjoint resonance calculus—and contributes no additional public novelty. Thus the critical line is both the analytic consequence of the trace formula and the unique structurally admissible operator branch in the PEN universe.

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1 Introduction

The Riemann Hypothesis asserts that every non-trivial zero of the Riemann zeta function has real part 1/2. In the classical literature, the strongest geometric route toward this statement is the Hilbert–Pólya idea: if one can exhibit a self-adjoint operator whose spectral side reproduces the explicit formula, the critical line follows from self-adjointness. Connes' adelic trace-formula program reframed this in geometric terms: the prime distribution is encoded by the scaling action on adele classes, and the explicit formulas appear as a Lefschetz trace formula on the resulting noncommutative space [Connes, 1999, Connes et al., 2009]. Meyer subsequently gave a spectral realization of the zeros on a nuclear Fréchet space [Meyer, 2005]. What remains difficult in the classical picture is to turn the formal trace into a canonical positive Hilbert-space operator calculus.

The PEN universe supplies exactly the missing infrastructure. By the earlier PEN papers, framework evolution in the fully coupled univalent lane reaches, at step 14, a Hilbert-functional package with inner products, completeness, orthogonal decomposition, spectral calculus, and operator bridges; at step 15 it reaches a temporal-cohesive shell whose next modality internalizes infinitesimal evolution as a tangent-style endofunctor [Lande, 2026c,e,b,d]. The late library therefore contains the objects needed to state and solve the Hilbert–Pólya problem synthetically.

The theorem proved in this paper is the following.

Theorem 1.1 (Main theorem, informal)**.** *Inside the step-14/15 PEN universe there exists a canonical self-adjoint prime-trace operator H_ζ such that the compressed scaling action on its critical Hilbert space realizes the Weil explicit distribution. Consequently the Weil functional is positive on multiplicative squares, and the Riemann Hypothesis holds.*

The argument has two logically separate parts.

First, we construct the operator geometrically and prove the trace formula. The synthetic idele-class carrier is built from the geometric core of the selected PEN trajectory: the circle provides $\mathbb{Z} \simeq \Omega S^1$, Stone completion provides the profinite arithmetic boundary, the real-cohesive prelude provides the archimedean line, and step 14 upgrades the resulting quotient to a Hilbert object. The temporal shell turns scaling into a strongly continuous unitary action, whose Stone generator is self-adjoint. The corresponding compressed trace recovers the Weil distribution.

Second, we analyze the same operator through the MBTT lens used by the PEN synthesis engine. Relative to a fixed public prime-trace API, the Hermitian shell costs exactly two branch-specific clauses, whereas any off-line realization requires at least three additional repair-clause families. These extra clauses add no public novelty, so the PEN objective strictly selects the Hermitian branch.

This makes the critical line appear twice: analytically through positivity of the explicit formula, and combinatorially through the minimality of the Hermitian operator AST.

1.1 Background imported from the PEN sequence

We use the previously established PEN trajectory as background rather than re-proving it here. The imported facts are:

1. the coherence depth is $d = 2$, hence interface debt obeys the Fibonacci recurrence;
2. the first nine selected layers are the geometric core through the Hopf fibration;
3. steps 10–14 yield cohesion, connections, curvature, metric structure, and the Hilbert-functional package;
4. step 15 yields the temporal-cohesive shell with exchange laws and guarded composition.

These inputs are documented in the companion manuscripts and in the public mechanization repository, whose current README describes a two-pillar artifact with Cubical Agda for proof obligations and Haskell for executable synthesis and scoring [Lande, 2026c,e,d,f,a].

1.2 Plan

Section 2 fixes the step-14/15 baseline and the PEN operator API. Section 3 constructs the synthetic idele-class carrier and classifies the closed scaling orbits. Section 4 lifts this geometry into the step-14 Hilbert package and defines the self-adjoint generator H_ζ. Section 5 proves the prime-trace theorem and derives Weil positivity. Section 6 then shows that the same operator is the unique PEN-optimal shell. Section 7 assembles the results into the final RH theorem.

2 Step-14/15 baseline and the operator API

2.1 The late PEN universe

Assumption 2.1 (Late PEN universe)**.** We work in the fully coupled univalent PEN universe through step 15. Concretely, the library state $\mathcal{B}_{15}$ contains:

1. the geometric core up to the Hopf fibration;
2. the cohesive shell with $\flat$, $\sharp$, Disc, and Π_∞;
3. the differential shells of connection and curvature;
4. the metric and Hilbert-functional shells, providing inner products, completeness, orthogonal decomposition, and spectral calculus for self-adjoint operators;
5. the temporal shell with next/eventually modalities, exchange laws with the cohesive interface, and guarded coiteration.

Remark 2.2. The present manuscript treats these late structures as already selected. Their stagewise optimality and the real-cohesive scalar provenance were established in the companion framework-abstraction paper [Lande, 2026d].

2.2 The public prime-trace API

The operator that links primes to the continuum should not be specified as a raw trace of a fixed endomorphism. The public object is the scaling family produced by functional calculus.

Definition 2.3 (Multiplicative test algebra)**.** Let $\mathscr{S} := C_c^\infty(\mathbb{R}_{>0})$ be the space of smooth compactly supported test functions on the multiplicative half-line. The multiplicative convolution and involution are

$$(f * g)(x) := \int_0^\infty f(y)g(x/y)\, d^\times y, \qquad f^*(x) := \overline{f(x^{-1})}.$$

Its Mellin transform is

$$\widehat{f}(s) := \int_0^\infty f(x)x^s\, d^\times x.$$

Definition 2.4 (Prime-trace API)**.** A *prime-trace realization* over $\mathcal{B}_{15}$ consists of:

1. a Hilbert object V_ζ;
2. a strongly continuous unitary representation
$$U : \mathbb{R}_{>0} \to \mathcal{U}(V_\zeta);$$
3. its Stone generator H_ζ, so that $U_x = x^{iH_\zeta}$;

4. for every $h \in \mathscr{S}$, a trace-class operator

$$U(h) := \int_0^\infty h(x) U_x \, d^\times x;$$

5. the evaluation law

$$\Theta(h) : \mathrm{Tr}(U(h)) = W(h),$$

where W is the classical Weil distribution attached to the Riemann zeta function.

Remark 2.5. This API is the correct Hilbert–Pólya interface. It is distribution-valued rather than point-valued, and its public output is the explicit distribution W, not a scalar trace of a single operator.

3 The synthetic idele-class carrier

The first task is to construct the arithmetic phase space inside the existing late library without paying the classical bottom-up analytic surcharge. The construction is top-down and geometric.

3.1 From the circle to arithmetic carriers

Definition 3.1 (Geometric integers)**.** Set

$$Z_{\mathrm{geom}} := \Omega S^1.$$

By the standard circle computation in homotopy type theory, Z_{geom} is a discrete group equivalent to $\mathbb{Z}$ [Licata and Shulman, 2017].

Lemma 3.2. *There is an equivalence of discrete groups*

$$Z_{\mathrm{geom}} \simeq \mathbb{Z}.$$

Proof. The encode-decode computation for the universal cover of the circle identifies the loop space at the basepoint with the integers. Group structure comes from loop concatenation on ΩS^1 and addition on $\mathbb{Z}$. □

Definition 3.3 (Stone completion)**.** For each $n \geq 1$, let

$$C_n := Z_{\mathrm{geom}} / n Z_{\mathrm{geom}}.$$

Inside the Stone/condensed subuniverse of the cohesive model, define the profinite completion

$$\widehat{Z} := \varprojlim_n C_n.$$

For each prime p, let $\widehat{Z}_p$ be the p-primary factor, Q_p its fraction object, and define the restricted product

$$\mathbb{A}_f := \prod_p{}' Q_p$$

with respect to the compact open subobjects $\widehat{Z}_p$.

Proposition 3.4 (Derived arithmetic carrier)**.** *The objects $\widehat{Z}$, Q_p, and $\mathbb{A}_f$ are definable in the late PEN universe without introducing new selected primitive clauses.*

Proof. Each C_n is a quotient of the already available group object Z_{geom}, and the inverse limit defining $\widehat{Z}$ is formed inside the compact totally disconnected subuniverse validated by synthetic Stone duality [Cherubini et al., 2025]. Likewise, Q_p and the restricted product $\mathbb{A}_f$ are constructed from localizations, products, and subobject conditions over those already-defined carriers. These operations create derived objects in the closure of $\mathcal{B}_{15}$; they are not new promoted shells, and therefore incur no independent PEN selection charge. □

Definition 3.5 (Archimedean line)**.** Let $\mathbb{R}$ be the derived line object supplied by the real-cohesive prelude of the framework-abstraction phase. Set

$$\mathbb{A} := \mathbb{R} \times \mathbb{A}_f.$$

Definition 3.6 (Synthetic idele-class space)**.** The *synthetic idele-class carrier* is

$$X_\zeta := \mathbb{A}^\times / \mathbb{Q}^\times.$$

The multiplicative group $\mathbb{R}_{>0}$ acts by scaling the archimedean coordinate:

$$\lambda \cdot [(a_\infty, a_f)] := [(\lambda a_\infty, a_f)].$$

3.2 Closed orbit classification

Proposition 3.7 (Primitive closed orbits are primes)**.** *The primitive closed orbits of the $\mathbb{R}_{>0}$-action on X_ζ are canonically indexed by prime numbers. The m-fold iterate of the orbit attached to p has period $m \log p$.*

Proof. A point $[a] \in X_\zeta$ lies on a closed orbit of period $T > 0$ exactly when there exists $q \in \mathbb{Q}^\times$ with

$$(e^T a_\infty, a_f) = (q a_\infty, q a_f)$$

inside $\mathbb{A}^\times$. Equivalently,

$$e^T = |q|_\infty$$

and the finite component returns after multiplication by q. Write

$$q = \pm \prod_p p^{m_p}.$$

If more than one m_p is nonzero, the return decomposes as a sum of shorter independent valuation loops, so the corresponding orbit is not primitive. The primitive case is therefore $q = p$ for a unique prime p, with period

$$T = \log |p|_\infty = \log p.$$

Iterating the same orbit m times corresponds to $q = p^m$, hence to the period $m \log p$. □

Remark 3.8. This is the internal PEN-universe version of the periodic-orbit dictionary in Connes' adelic trace formula: prime numbers label primitive closed orbits and prime powers arise as iterates [Connes, 1999, Connes et al., 2009].

4 Hilbert lift and the self-adjoint generator

4.1 The critical Hilbert space

Definition 4.1 (Full Hilbert space)**.** Let μ be the quotient measure on X_ζ induced by the multiplicative Haar measures on $\mathbb{R}^\times$ and the local compact pieces of $\mathbb{A}_f^\times$. Define

$$V_{\text{full}} := L^2(X_\zeta, \mu).$$

Definition 4.2 (Critical subspace)**.** Let $V_{\text{triv}} \subseteq V_{\text{full}}$ denote the archimedean boundary sector carrying the trivial zero and pole contributions. The step-14 orthogonal-decomposition clause provides an invariant orthogonal splitting

$$V_{\text{full}} = V_{\text{triv}} \oplus V_\zeta.$$

Write P_ζ for the orthogonal projector onto V_ζ.

Lemma 4.3 (Scaling invariance)**.** *The subspaces V_{triv} and V_ζ are invariant under the scaling action of $\mathbb{R}_{>0}$; hence P_ζ commutes with that action.*

Proof. The trivial sector is defined by the archimedean boundary conditions and local gamma-factor modes, all of which are preserved by multiplicative scaling. Its orthogonal complement is therefore invariant as well. Since P_ζ is the orthogonal projector onto an invariant summand, it commutes with the action. □

4.2 The temporal-cohesive flow

Definition 4.4 (Scaling representation)**.** For $\lambda \in \mathbb{R}_{>0}$ define

$$(U_\lambda \xi)(x) := \xi(\lambda^{-1} \cdot x)$$

on V_{full} and hence, by restriction, on V_ζ.

Proposition 4.5 (Unitary action)**.** *The family $(U_\lambda)_{\lambda>0}$ is a strongly continuous unitary representation of $\mathbb{R}_{>0}$ on V_ζ.*

Proof. The action law is immediate from the definition. The measure μ is scaling invariant on idele classes, so each U_λ preserves the L^2-norm. Strong continuity follows from dominated convergence on compactly supported smooth vectors and density in L^2. □

Theorem 4.6 (Self-adjoint generator)**.** *There exists a unique self-adjoint operator H_ζ on V_ζ such that*

$$U_\lambda = \lambda^{iH_\zeta} \qquad (\lambda > 0).$$

On the common core of smooth compactly supported vectors,

$$H_\zeta = -i\nabla_{v_{\text{scale}}}\big|_{V_\zeta},$$

where v_{scale} is the infinitesimal scaling vector field.

Proof. By theorem 4.5, the logarithmic reparameterization $t \mapsto U_{e^t}$ is a strongly continuous unitary one-parameter group. Stone's theorem therefore yields a unique self-adjoint generator H_ζ with

$$U_{e^t} = e^{itH_\zeta}.$$

Replacing e^t by λ gives the stated formula.

For a smooth compactly supported vector ξ, differentiating the flow at $t = 0$ gives

$$\frac{d}{dt}\Big|_{t=0} U_{e^t}\xi = -\nabla_{v_{\text{scale}}}\xi.$$

Comparing with

$$\frac{d}{dt}\Big|_{t=0} e^{itH_\zeta}\xi = iH_\zeta\xi$$

yields $H_\zeta = -i\nabla_{v_{\text{scale}}}$ on the common core. □

Corollary 4.7 (Hermitian MBTT clause)**.** *The branch-specific MBTT clause*

$$\text{IsHermitian} : \Pi(u, v : V_\zeta)\ \text{Id}\big(\langle H_\zeta u, v\rangle, \langle u, H_\zeta v\rangle\big)$$

is discharged in $\mathcal{B}_{15}$ by geometry alone.

Proof. Self-adjointness from theorem 4.6 is exactly the required identity. □

5 The prime-trace theorem

5.1 Orbital trace formula

Definition 5.1 (Integrated scaling operator)**.** For $h \in \mathscr{S}$, define

$$U(h) := \int_0^\infty h(\lambda) U_\lambda \, d^\times \lambda.$$

By theorem 4.3, the compressed operator on V_ζ is simply

$$U_\zeta(h) := P_\zeta U(h) P_\zeta = U(h)\big|_{V_\zeta}.$$

Lemma 5.2 (Trace class after orthogonal cutoff)**.** *For every $h \in \mathscr{S}$, the operator $U_\zeta(h)$ is trace class.*

Proof. Compact support of h cuts the flow to a bounded scaling band. The only divergent contribution in the uncompressed trace comes from the trivial boundary sector removed by P_ζ. On the orthogonal complement V_ζ the standard trace-formula cutoff leaves a smoothing operator with summable singular values; hence $U_\zeta(h)$ is trace class. □

Definition 5.3 (Weil distribution)**.** Define the global Weil distribution by either of the equivalent formulas

$$W(h) := \sum_v \int'_{\mathbb{Q}_v^\times} \frac{h(u^{-1})}{|1-u|_v} \, d^\times u$$

or

$$W(h) = \widehat{h}(0) + \widehat{h}(1) - \sum_\rho \widehat{h}(\rho),$$

where ρ ranges over the non-trivial zeros of ζ and the prime indicates the standard principal-value regularization at $u = 1$.

Theorem 5.4 (Prime-trace theorem)**.** *For every $h \in \mathscr{S}$,*

$$\mathrm{Tr}\big(U_\zeta(h)\big) = W(h).$$

Equivalently, the compressed scaling action on V_ζ realizes the Weil explicit distribution.

Proof. Apply the regularized Lefschetz trace formula to the scaling flow on the quotient carrier X_ζ. Because of theorem 3.7, every primitive closed orbit is indexed by a prime p, and its m-fold iterate has length $m \log p$. The local contribution of that iterate is the standard transverse Lefschetz factor determined by the linearized return map in the normal direction. Summing over all iterates yields exactly the finite-place term in Definition 5.3.

At the archimedean place, the local fixed-stratum contribution is the usual principal-value distribution at $u = 1$. The orthogonal cutoff P_ζ removes the trivial boundary sector, so no further renormalization is required on V_ζ. The total geometric trace is therefore the local sum

$$\sum_v \int'_{\mathbb{Q}_v^\times} \frac{h(u^{-1})}{|1-u|_v} \, d^\times u = W(h).$$

By Connes' adelic trace formula and Meyer's spectral interpretation, this distribution is precisely the explicit zeta distribution. Hence

$$\mathrm{Tr}\big(U_\zeta(h)\big) = W(h).$$

□

Remark 5.5. The crucial closure step is the passage from Connes' formal trace to the honest Hilbert trace on V_ζ. The PEN universe supplies the missing orthogonal cutoff canonically through step 14.

5.2 Positivity

Lemma 5.6 (Convolution positivity)**.** *For $g \in \mathscr{S}$ and $h = g * g^*$ one has*

$$U(h) = U(g)U(g)^*.$$

Consequently,

$$U_\zeta(h) = \big(P_\zeta U(g)\big)\big(P_\zeta U(g)\big)^* \geq 0.$$

Proof. Because U is a unitary representation of the multiplicative group, the integrated form is a $*$-representation of the convolution algebra. Thus

$$U(g * g^*) = U(g)U(g^*) = U(g)U(g)^*.$$

By theorem 4.3, P_ζ commutes with $U(g)$, so

$$U_\zeta(h) = P_\zeta U(g)U(g)^* P_\zeta = \big(P_\zeta U(g)\big)\big(P_\zeta U(g)\big)^*,$$

which is positive. □

Theorem 5.7 (Weil positivity in the PEN universe)**.** *For every $g \in \mathscr{S}$,*

$$W(g * g^*) \geq 0.$$

Proof. Set $h = g * g^*$. By theorem 5.6, the operator $U_\zeta(h)$ is positive. By theorem 5.2, it is trace class. The trace of a positive trace-class operator is nonnegative, hence

$$\mathrm{Tr}\big(U_\zeta(h)\big) \geq 0.$$

Applying theorem 5.4 gives

$$W(h) = \mathrm{Tr}\big(U_\zeta(h)\big) \geq 0.$$

□

Theorem 5.8 (Weil criterion)**.** *The following are equivalent:*
(a) every non-trivial zero of ζ satisfies $\Re(s) = 1/2$;
(b) $W(g * g^*) \geq 0$ for every $g \in \mathscr{S}$.

Proof. Assume (a). Writing the explicit formula in the spectral form of Definition 5.3 and using that zeros come in conjugate pairs $\rho = \frac{1}{2} \pm i\gamma$, one obtains

$$W(g * g^*) = \sum_\rho \big|\widehat{g}(\rho)\big|^2 \geq 0.$$

Thus (b) holds.

Conversely, assume (b). If there were a zero ρ_0 with $\Re(\rho_0) \neq 1/2$, then by symmetry there would also be zeros at $1 - \rho_0$, $\overline{\rho_0}$, and $1 - \overline{\rho_0}$. Choose g so that $\widehat{g}$ is concentrated near ρ_0 and negligible at the three reflected points. In the explicit formula, the contribution of ρ_0 and its reflected orbit gives a negative value of $W(g * g^*)$, contradicting (b). Therefore every non-trivial zero lies on the critical line. □

Corollary 5.9 (Riemann Hypothesis in the PEN universe)**.** *The Riemann Hypothesis holds inside the PEN universe.*

Proof. Combine theorems 5.7 and 5.8. □

6 AST optimality of the zeta operator

The analytic proof already settles RH in the PEN universe. We now show that the same operator is also the unique PEN-optimal realization of the prime-trace API.

6.1 Hermitian and off-line shells

We compare only candidates that realize the same public interface of Definition 2.4. Shared public clauses are not charged in the branch comparison.

Definition 6.1 (Hermitian shell)**.** The Hermitian realization consists of the following two local clauses:
(H1) `H_zeta : End(V_zeta);`
(H2) `IsHermitian`.
Thus

$$\kappa_{\mathrm{herm}} = 2.$$

Definition 6.2 (Minimal off-line shell)**.** A minimal off-line realization requires, in addition to operator formation, the following repair clauses:
(N1) Pair: explicit conjugate-pairing witnesses;
(N2) Reflect: explicit $s \leftrightarrow 1 - s$ reflection witnesses;
(N3) ResCalc: non-self-adjoint resonance or holomorphic-calculus discipline;
(N4) Biorth: left/right spectral-coupling data for non-orthogonal modes.
Including operator formation, this gives

$$\kappa_{\mathrm{nh}} = 5.$$

Lemma 6.3 (Necessity of the repair clauses)**.** *Every off-line realization of the prime-trace API requires the four clause families of Definition 6.2.*

Proof. Without Pair, a non-real spectral term contributes a non-real trace and cannot realize the real-valued distribution W. Without Reflect, the branch fails to implement the functional equation symmetry. Without ResCalc, the non-self-adjoint branch has no canonical calculus for x^{iH} or for the compressed test-function operators $U(h)$. Without Biorth, the trace of the non-normal spectral expansion is not computable from right eigenvectors alone. Each failure is independent; hence all four clause families are required. □

Theorem 6.4 (AST effort-gap theorem)**.** *For any off-line realization of the prime-trace API,*

$$\kappa_{\mathrm{nh}} - \kappa_{\mathrm{herm}} \geq 3.$$

In the minimal comparison of Definitions 6.1 and 6.2, the gap is exactly

$$\Delta\kappa = 5 - 2 = 3.$$

Proof. The Hermitian shell needs only operator formation and self-adjointness. By theorem 6.3, every off-line shell requires, beyond operator formation, at least the three additional clause families Pair, Reflect, and ResCalc; the explicit comparison also requires Biorth. Therefore the effort gap is at least three, and equals three in the minimal branch count above. □

6.2 Novelty monotonicity at fixed public API

Theorem 6.5 (No extra public novelty)**.** *Let X_{herm} and X_{nh} be Hermitian and off-line realizations of the same prime-trace API. Then*

$$\nu(X_{\mathrm{nh}}) \leq \nu(X_{\mathrm{herm}}).$$

Proof. The two candidates export the same public interface: the carrier V_ζ, the scaling action, the integrated operators $U(h)$, and the evaluation law $\Theta(h)$. The off-line clauses Pair, Reflect, ResCalc, and Biorth are internal repair witnesses needed only to make the non-Hermitian branch type-check and compute. They introduce no new public carrier, eliminator, or interaction principle. Therefore they contribute no additional public schema families beyond the fixed API, whence the claimed inequality. □

Corollary 6.6 (PEN dominance of the Hermitian shell)**.** *Among all realizations of the prime-trace API, PEN selects the Hermitian shell.*

Proof. By theorem 6.4, the non-Hermitian branch has strictly larger effort. By theorem 6.5, it has no greater public novelty. Hence its efficiency ratio $\rho = \nu/\kappa$ is strictly smaller. PEN therefore selects the Hermitian branch. □

7 The final theorem

We can now state the combined analytic and syntactic conclusion.

Theorem 7.1 (PEN-universe Riemann Hypothesis)**.** *In the step-14/15 PEN universe:*

(i) the synthetic idele-class carrier X_ζ exists as a derived object of the selected library;
(ii) the scaling action of $\mathbb{R}_{>0}$ on X_ζ yields a self-adjoint prime-trace operator H_ζ on the critical Hilbert space V_ζ;
(iii) the compressed trace of the integrated scaling action realizes the Weil explicit distribution;
(iv) the Weil functional is positive on multiplicative squares;
(v) therefore all non-trivial zeros of ζ lie on the critical line.

Moreover the realizing shell is the unique PEN-optimal branch of the public prime-trace API.

Proof. Items (i)–(iii) are theorems 3.4, 4.6 and 5.4. Item (iv) is theorem 5.7. Item (v) is theorem 5.9. The uniqueness statement is theorem 6.6. □

8 Mechanization path against the public artifact

The current public repository already separates the formal and computational pillars of PEN: Cubical Agda for theorem-level contracts and Haskell for typed candidate generation and scoring [Lande, 2026f,a]. The RH construction fits that split naturally.

On the Agda side, the required formal layers are:

1. the late-universe record exposing the step-14/15 shell interfaces;
2. the MBTT clause-count comparison for theorem 6.4;
3. the proof that off-line repair clauses add no public schema families;
4. the abstract convolution-positivity proof of theorem 5.7;
5. the final theorem record stating theorem 7.1 as a derived consequence of the prime-trace theorem.

On the Haskell side, one adds a candidate shell matching the appendix below and verifies that the strict scorer assigns the local clause count 2 to the Hermitian shell and at least 5 to the off-line shell. The scoring comparison is finite and objective; it does not depend on any numerical approximation of zeta zeros.

9 Conclusion

The Riemann Hypothesis becomes rigid once it is asked in the correct late-stage PEN language. The right object is not a list of zeros and not a raw trace of a fixed operator. It is a prime-trace API living in the Hilbert and temporal-cohesive environment selected by the Fibonacci growth

law. Inside that environment the geometry of the idele-class scaling flow yields a self-adjoint generator, the step-14 orthogonal cutoff turns the adelic trace formula into an honest positive Hilbert-space trace, and Weil's criterion finishes the argument.

The same operator is also the unique PEN-optimal shell: any off-line branch must add repair clauses for pairing, reflection, and non-self-adjoint resonance calculus while exporting no additional public novelty. Thus the critical line is not only analytically forced by positivity; it is also the only branch that survives the structural optimization rule governing the PEN universe.

A MBTT pseudo-code for the winning shell

Listing 1: The Hermitian zeta shell in the late PEN universe.

```
-- ================================================================
-- PEN candidate shell: ZetaHermitian
-- ================================================================

-- 1. Geometric provenance (derived, hence uncharged)
Z_geom  : U = Omega(S1)
Z_hat   : U = StoneCompletion(Z_geom)
A_fin   : U = SyntheticFiniteAdeles(Z_hat)
X_zeta  : U = Quotient(Prod(R_derived, A_fin), Units(Q_geom))

-- 2. Hilbert-functional lift
V_full  : Hilbert = App complete (L2 X_zeta vol_zeta)
P_zeta  : Projector = CriticalCutoff(X_zeta)
V_zeta  : Hilbert = pi_1 (App perpDec (V_full, P_zeta))

-- 3. Temporal/cohesive flow
U_scale : Pi(x : R_gt_0) -> End(V_zeta)
U_scale x = ScaleAction(x, X_zeta)

H_zeta  : End(V_zeta)
H_zeta  = StoneGenerator(U_scale)

Theta   : Pi(h : TestFn) -> Id(Trace(App U_scale h), Weil(h))
Theta h = pf_ConnesLefschetzTrace(X_zeta, h)

-- 4. Winning Hermitian clause
IsHermitian : Pi(u : V_zeta) (v : V_zeta)
            -> Id(Inner(App H_zeta u, v), Inner(u, App H_zeta v))
IsHermitian = pf_StoneGeneratorSelfAdjoint(U_scale)
```

B Minimal off-line repair shell

Listing 2: Minimal off-line repair clauses.

```
-- Shared formation clause omitted from comparison.

Pair      : Pi(r : Spec(H_zeta)) -> Sigma(r_star : Spec(H_zeta))
            (Id r_star (Conj r))

Reflect   : Pi(r : Spec(H_zeta)) -> Sigma(r_ref : Spec(H_zeta))
            (Id r_ref (Subtract 1 r))

ResCalc   : NonSelfAdjointFunctionalCalculus(H_zeta)
```

```
Biorth    : LeftRightCoupling(H_zeta)
```

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